

On De-anonymization of Single Tweet Messages

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ABSTRACT

In this work, we address the question of whether the authorship of a single tweet can be successfully identified (and in a mixed set with other authors). Here, we present a new authorship identification scheme, which is useful in detecting authorship of short texts such as tweets, in case where only single messages are available. Our authorship identification scheme relies on selecting features that work for the special setting and combine them in order to obtain a better accuracy. This technique demonstrates significant results through our experiments. Our results can be used to detect authors of illegitimate tweets, fake tweets in a Twitter account or break the privacy of a multi-user account by showing the authors who participate in it.

KEYWORDS

Online social networks; Twitter; Stylometry; Privacy; Anonymity

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1 INTRODUCTION

Application of authorship attribution technique has been expanded to domains beyond regular texts such as reports, excerpts of books, to computer-facilitated texts (including source codes, emails, blogs, posts on social networks or forums, reviews and comments). Twitter messages (tweets) are very short (less than 140 characters), which poses difficulty for authorship identification algorithms that rely on substantial pieces of texts. This is particularly true when authorship of a single tweet is considered. Standard stylistic markers such as lexical richness, frequency of function words, or other syntactic measures; which are known to perform well with longer standard language texts, perform much worse with such short texts. In this work, we address the question of identifying the author of very short texts. There are various motivations for this study. In underground world, Twitter can be used to send offensive and illegitimate tweets, as seen in the news [14] [21]. Sometimes the suspects of even single tweet messages that contain threats need to be verified

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immediately. A direct application of our work is to detect the owner of these tweets, or to narrow down the number of suspects. Another important motivation comes from scenarios where account is broken into, and fraudulent tweets are sent on behalf of the account holder. Alternatively, the account owner would like to deny a tweet and claims that the account is broken into. A second scenario is when a group of individuals use the same Twitter account with the goal of preserving their privacy. Our goal is to show that with access to sufficient amount of data, one can detect the ownership of single tweets.

The main contributions of this work are as follows:

- We provide an analysis of the previous works and discuss the potential drawbacks of the works.
- A new authorship attribution scheme to detect the authorship of an anonymous tweet. Our results show that accuracy significantly improves when we detect authorship of single tweets within tweet domain using our algorithm. We can narrow down the number of suspects to 3 from a set of 50 authors with only 100 tweets in the training set. The accuracy is nearly 95% when we detect authorship in a set of 2 authors. Our work doubles the accuracy of the current work.

Outline: Section 2 is an overview of the related works. An analysis of tweets' characteristics and problem statement are presented in Section 3 and 4. We analyze the previous work and describe our algorithm in Section 5. Section 6 presents our experimental results. Section 7 is the summary of our paper.

2 RELATED WORKS

In other domains of texts such as literature, reports, blogs, etc., stylometry is considered as a powerful tool to perform authorship attribution over documents. For a document of at least 250 words, traditional authorship attribution reports good accuracy on attributing correct author to it [2], even in the Internet-like scale [12]. Authorship attribution techniques then can be utilized to identify anonymous writings in these domains with accuracy above 90% [19]. This technique hence can also be used to detect the linkability of accounts across websites.

However, with micro-messages such as tweets where the length of texts are micro, the traditional authorship attribution techniques cannot provide good accuracy when they are applied to this domain. In the literature, there are a number of works that address authorship detection and account-linkability within tweet domain.

Noteable results have been achieved in authorship detection of tweets. Experimenting on groups of 75 tweets, in a scale of 3 users, in [17], Silva et al. could achieve 50% recall at 90% precision rate. Non-traditional stylometry features were used, such as user mentions, number of hashtags. Mixes of traditional and non-traditional features are used in [7] by Layton et al., the authors could detect authorship of 120 tweets with accuracy 72% amongst

50 authors, and in [4] Bhargava's algorithm achieved 80% accuracy when identifying authorship of 200 tweets in a set of 20 authors.

In terms of scale, in [9] (Mishari et al.), the authors attempted to link sets of 5, 10,... to 100 tweets with a large set of identified Twitter accounts. Naive Bayes classifier was trained on a set of character n -grams ($n = 1, 2$). The linkability result is approximately 50%. When hashtags are included, it helps improve the result.

Recently, in cross-domain setting, Overdorf et al. in [3] and [15] presented their algorithms to identify if a group of tweet feeds (500 words) and a group of blogs/Reddit comments (4500 words) are from the same author.

In these works, features are not designed for the specific setting when only single tweets are available for detecting authorship. These features work only when a set of tweets are available for training and testing phase. According to our test, both character n -grams ($n=1,2$) and non-traditional features, when they are treated as independent ones, result in insignificant accuracy.

In [16], the problem of detecting authorship of a single tweet feed was mentioned, we review this paper in details in section 5.1.

3 TWEET CHARACTERISTICS

Messages on Twitter (which are called "tweets") are restricted to 140 characters. Users heavily use self-defined hashtags and user mentions in tweets. Within the short length, they invented many techniques to expand the semantics of the short messages. For example, when posting external urls, users often use url shortening services (e.g., <http://www.bit.ly> or <http://goo.le> service). Tweeters also tend to generate more abbreviations, misspellings, ... and change their styles such as omitting words or not following syntax rules. In our setting, we keep these characteristics in authors' messages and similar writing styles are captured by character 4-grams. Treating these characteristics as independent feature types would result in low recall rates as user models in the testing phase are constructed from single messages.

4 PROBLEM STATEMENT

Suppose there is a mixed set of tweets, and there is a set of suspects that can be narrowed down to $N > 0$ authors: $\{A_1, A_2, \dots, A_N\}$.

Assume that each candidate A_i ($1 \leq i \leq N$) have another public set of tweets. The problem becomes given A_i 's training tweet set, identify the authorship of *single tweets* t in the mixed set.

5 OUR WORK

5.1 Analysis of R. Schwartz et al.'s [16]

There are two main questions in authorship detection of single tweets: what possible feature sets are and how to refine these feature sets such that they contain only the most effective features. In [16], the problem of detecting authorship of a single tweet feed was mentioned with a selective framework for the setting, we analyze the previous work based on these two criteria.

Feature sets. The used feature sets include word n -grams and character 4-grams.

Character n -grams. 4-grams as well as n -grams characters are strong candidates for representing an author's writing style as they are known to be tolerant with typos (for example, both Disney and

Disney can produce the 4-gram 'Disn'), unusual uses of punctuations, whitespaces or word orders. These are common informal styles found in social media like Twitter. *In our experiments, we extend the range of character n -grams to $n = 3, 4$ to maximize the capture strength of character n -grams.*

Word n -grams. words such as function words and some common words that are independent of contexts could be used repetitively in writings by the same author. Beside function words, there is no pre-defined list of other context-independent words that are used by one person (for example, one could use 'go', 'see', ... multiple times is his or her writings in different contexts and they are not listed as function words). These words reflect one's writing style such as casual or formal, which does not necessarily tie to a specific topic. We present more examples of these words and a list of function words in section Appendix. To capture this part of author's writing style in [16], the authors considered word n -grams with $2 \leq n \leq 5$. However, it is more likely that smaller size word phases (n is small) are repeated by Twitter users in this setting where there are only single tweets for the identification task. *Therefore, we consider word n -grams with $n = 1, 2$.* Although word phases are not strong candidates for representing writing style, however, if exists, this part of author's writing style can be added and enhance the current existing feature sets such as character n -grams.

Due to the short length of single tweets, traditional features such as lexical richness, sentence length, sentence count, and other stylometric distributions cannot be extracted. On the other hand, non-traditional features are kept in the tweet's content as analyzed in Section 3 to strengthen the feature sets.

Feature selection. The algorithm uses k -signature concept - it selects features that appear in at least k percents of training data of author A , but do not appear in others authors'.

In our opinions, this does not capture the whole of an author's style, as it is common that a feature that represents an author could also be present in other authors' writings with different patterns. As the result, k -signature may miss important ones that identify one author.

We propose a new approach to select features that represent one author's writing for authorship detection of single tweets as below.

5.2 Our proposed algorithm

To refine our feature set such that it includes only the most distinguishing features of an author, we use mutual information between a feature and an author. Mutual information reflects the occurrence and co-occurrence patterns of a feature and an author, thus it reflects the relationship between them, and can be used to measure the importance of that feature in representing the author. The higher mutual information value means the author and the feature are more likely to appear together. Only the most important features are included in the final feature set.

k -signature is a special case of mutual information where it ranks the features appear only in one author's writings which results in missing other distinguishing features. Our implementation of mutual information $I(A_k, f_i)$ between a feature f_i and an author A_k is described as following.

Calculating $I(A_k, f_i)$.

Output: top ℓ features f_i to include in the feature set F^{A_k} of A_k ($k = \{1, 2, \dots, N\}$).

We calculate the mutual information between each feature f_i extracted from author A_k and A_k . First we construct a corpus C of tweets belong to $A_1, A_2, \dots, A_N$.

Let X_{f_i} and X_{A_k} denote two binary variables that represent the presence of f_i in a tweet or A_k is the author of a tweet. The mutual information between f_i and A_k is calculated as :

$$I(X_{f_i}, X_{A_k}) = H(X_{f_i}) + H(X_{A_k}) - H(X_{f_i}, X_{A_k})$$

$p(X_{f_i} = 1)$ is the probability that f_i appears in a tweet, $p(X_{A_k} = 1)$ is the probability that A_k is the author of a tweet, and $p(X_{f_i} = 1, X_{A_k} = 1)$ is the probability that f_i presents in a tweet of A_k , where probabilities are calculated as relative frequencies:

$$\begin{aligned} p(X_{f_i} = 1, X_{A_k} = 1) &= \frac{N_{f_i \wedge A_k}}{N_C} \\ p(X_{f_i} = 1) &= \frac{N_{f_i}}{N_C} \\ p(X_{A_k} = 1) &= \frac{N_{A_k}}{N_C} \end{aligned}$$

Here N_C is the number of tweets in the corpus, C , $N_{f_i \wedge A_k}$ is the number of tweets in C that contain f_i and belong to A_k , and N_{f_i} , N_{A_k} are the numbers of tweets in C contain f_i , and written by A_k separately.

Mutual information of features are ranked in descending order, and the top ℓ features are selected to F^{A_k} .

Algorithm 1 describes our authorship identification algorithm, where δ is the selected classifier. Pre-processing step includes removing retweets and repeated tweets. The algorithm extracts and refines feature sets for each author. These feature sets are then combined together to represent one author's writing. Features extracted from tweets in the test phase are not further refined due to the small sizes of the sets.

Classification selection. There is a wide range of classifiers to perform text classification. Weka [5], a tool set used for text classification research, provides most of these classifiers. The tool set includes implementations for SVM, Naive Bayes, Neural Network, Decision Tree and so on. We use k -fold cross-validation method with $k = 10$ to evaluate the performance of these classifiers. The outperforming classifier is SMO SVM and is selected as δ in our authorship identification algorithm. SVM is also an efficient algorithm used in literature to perform text and other high dimensional classification.

5.3 Measurements

For identification of tweets, beside accuracy which measures the ratio of correctly identified tweets, the following metrics are useful to evaluate the algorithm:

- Precision ($= \frac{tp}{tp+fp}$): the fraction of correctly identified tweets among all tweets that are identified as class A in the testing set. A higher precision means fewer tweets are misidentified as A 's samples. This is equivalent to the true positive rate.

- Recall ($= \frac{tp}{tp+fn}$): the fraction of correctly identified tweets of all tweets belong to an author.

In the above formulas, tp : true positive cases, tn : true negative cases, fp : false positive cases, fn : false negative cases.

Input : single tweet t , author set $\mathcal{A} = \{A_1, A_2, \dots, A_N\}$, training data from N authors $\mathcal{D} = \{D_1, D_2, \dots, D_N\}$

Output: authorship of t

- 1 Preprocess $t, D_1, D_2, \dots, D_N$
- 2 /* Extract feature subset j for every author A_i :
*/
- 3 $FeatureExtractor(D_i) \leftarrow F_j^{A_i}$
- 4 Extract each feature subset j for $t: F_j^t$
- 5 Combine all subsets:
- 6 $F^{A_i} \equiv \bigcup_j F_j^{A_i}, F^t \equiv \bigcup_j F_j^t$
- 7 Form the training set: $F^A \equiv \bigcup_i F^{A_i}$
- 8 /* Identification: */
- 9 $Author = \delta(F^t, F^A)$

Algorithm 1: Tweet authorship identification

5.4 Computational cost

The computational cost of our algorithm except classification task is feature extraction cost, that is $O(m|C|)$, where m is the total number of features extracted from an author's writings, and $|C|$ is the size of the corpus.

6 EXPERIMENTS

6.1 Settings

We collected tweets from actual accounts who have from 300 to thousands of tweets. In total, we collected 100 accounts. Most recent tweets are gathered using Twitter4J APIs [6], and only tweets with at least 10 words are considered (links, mentions, and hashtags are counted as words). In experiments described below, each dataset is divided into training and test sets. An author's training data include a set of tweets. Each test document contains a single tweet.

We define a single test as follows. We construct 100 sets of mixed tweets by selecting tweets from N random users. We start with $N = 50$ users. We run the algorithm to detect the authorship of each tweet in the set.

- Experiment with different feature types: we repeat experiments with our selected features from Section 5 to measure their robustness. A better feature type is the one obtains higher accuracy in authorship detection of single tweets.
- The algorithms are evaluated through different numbers of candidates (N ranges from 2 to 50) and different training data sizes. We aim at optimal settings, thus we test with as small number of tweets in the training set as possible (from 15 tweets and upward).
- The classifier uses a threshold of confidence to decide if a test belongs to a class. In some applications such as to show that someone has broken into an account and sent fake tweets, the true positive rate is more important than the recall rate. On the other hand, when

the application needs to search for as many tweets written by a specific author as possible, the threshold is decreased to increase the recall rate. We experiment to demonstrate the precision and recall trade-off.

6.2 Results and analysis

Figure 1 illustrates the effectiveness of feature sets with different number of tweets in the training set and in a pool of 50 users (the random accuracy baseline is 2%). Character 4-grams provide the best accuracy which is nearly 30% with 100 tweets and the accuracy is improved when the number of training tweets increases. In addition, the other feature sets also give good results.

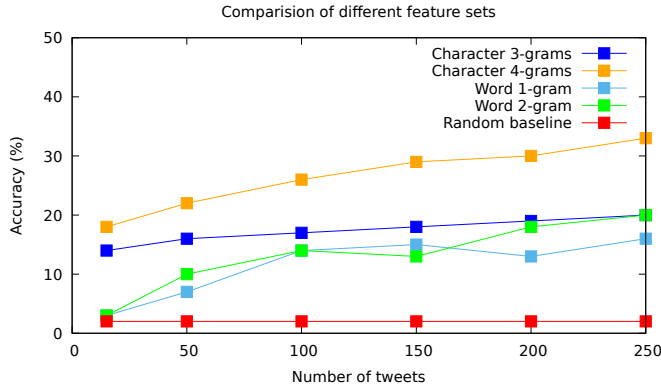


Figure 1: Accuracy of detecting authorship of single tweets with different feature sets. The random baseline accuracy is 2%.

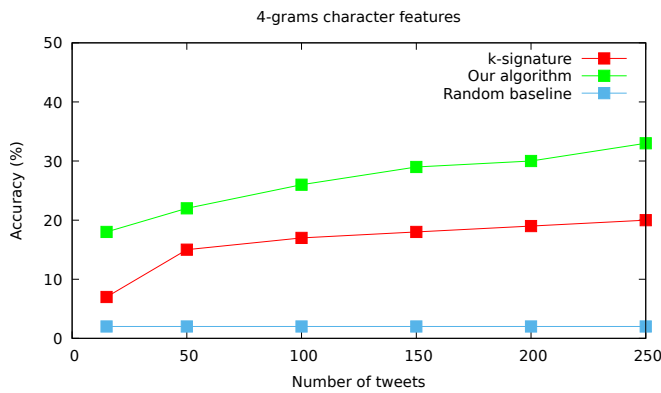


Figure 2: Accuracy of authorship detection of single tweets with different training set sizes using 4-gram characters by our algorithm and k -signature. The random baseline accuracy is 2%.

With a set of 50 candidates, and with different number of tweets in the training set, we tested both our algorithm and k -signature [16]. It is demonstrated in Figure 2, that our algorithm out-performed the k -signature with nearly doubled accuracy at every set of training data. Given as few as 15 tweets, we can detect the authorship

of single tweets in a pool of 50 authors with accuracy 18% while the random chance is 2%, which is equivalent to narrowing down to 5 candidates in a set of 50 authors on average if we consider the real author appears in the top 5 of a list ranked by classification confidence.

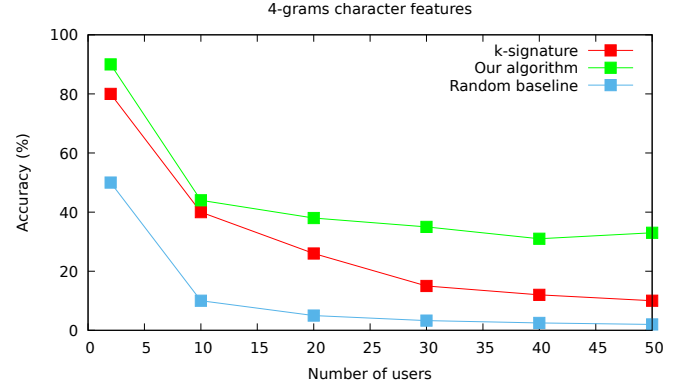


Figure 3: Accuracy of authorship detection of single tweets among different number of candidates using 4-gram characters by our algorithm and k -signature.

The accuracy increases when the number of candidates decreases. Our algorithm could achieve above 90% accuracy when separating tweets from a set of two authors (k -signature achieved approximately 70%). Figure 3 demonstrates the results of the algorithms with 250 tweets in the training set. This work shows that our approach becomes better with larger number of candidates.

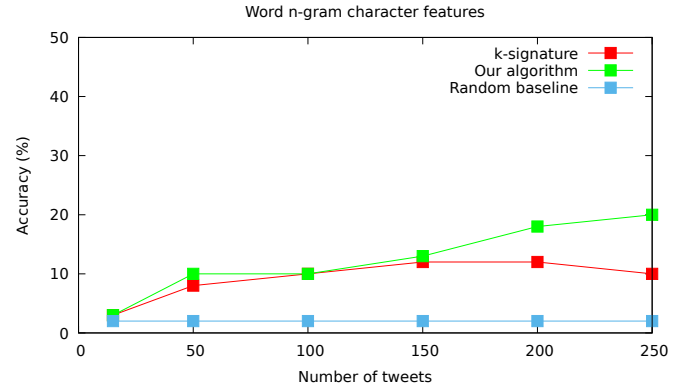


Figure 4: Accuracy of detecting authorship of single tweets with different training set sizes using word n -grams ($n = 2$) by our algorithm and k -signature. The random baseline accuracy is 2%.

Tested on word n -grams (with $n=2$), from 250 tweets per training set, we can get 20% accuracy when detecting authors of single tweets in a pool of 50 candidates. Figures 4 and 5 illustrate the effectiveness of word n -grams using our algorithm and k -signature. Our algorithm constantly achieved better accuracy compared to the previous work by up to 10%.

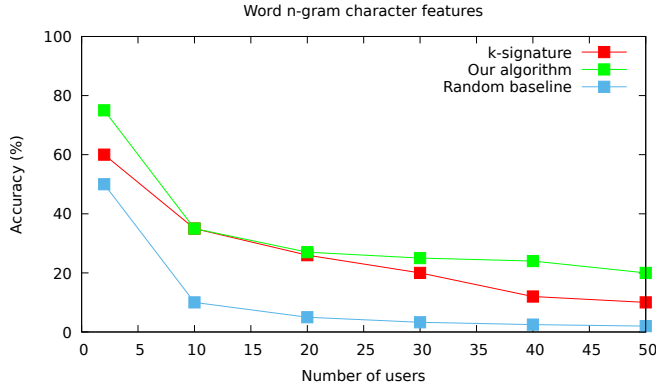


Figure 5: Accuracy of detecting authorship of single tweets with different number of candidate authors using word n -grams ($n = 2$) by our algorithm and k -signature (with 250 tweets per training set).

The trade-off between the recall and precision rate of a class is depicted in Figure 6. To obtain a high recall rate, for example, if almost all tweets written by A must be returned, we can decrease the classifier's threshold. The returned set may contain tweets from other authors which is 33% precision. In other words, to increase the response rate to a class, we can only narrow down the number of candidates. The threshold can be set high so that the returned set is not mixed or the precision rate is 100%, and the algorithm could response 60% of A 's tweets.

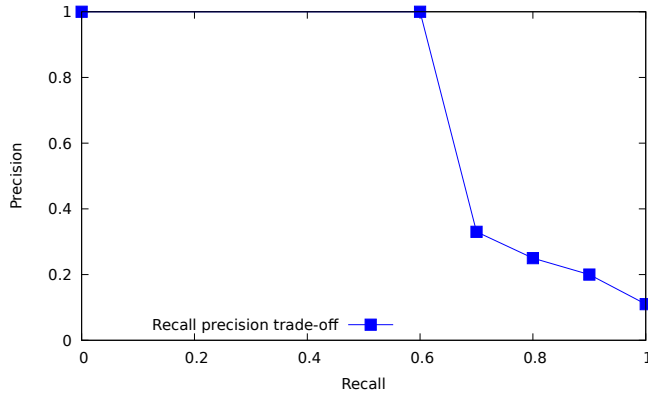


Figure 6: Recall and precision values of a class when the training size is 250 tweets and the number of authors is 10.

Discussions. In our experiments, we set the number of features ℓ (Section 5.2) for each feature set is 100 to 200 (about 2% of the set size) and obtain the best results. Although SVM classifier can deal with vectors with a higher number of dimensions, we found that increasing feature set size does not help increase the overall accuracy.

With the same computational cost, our algorithm outperforms [16], as shown above. These improvements are explained by the choice of our feature selection method.

We found that character 3-grams and word 1-gram also work well in this setting and when any of them (including word 2-grams) combines with character 4-grams, the accuracy is increased by approximately 3%. On the other hand, word n -grams with $n > 2$ give performances not much significant than a random chance.

We note that in [16], the authors further applied "flexible patterns" on their word n -grams. "Flexible patterns" are sequences of generalized forms of words (high frequency words and content words). However this technique only helped improve the overall accuracy by less than 3%.

6.3 Writings on Twitter

We analyzed the cases where the authorship of tweets that the algorithm could not detect (i.e., false negatives) and could detect with high accuracies, we have following statistics about authors' writing. First, for every random author, a tweet can have from 0 to 3 common 4-gram characters with that author. Thus, there are chances that the tweet has more common features with other authors.

On the other hand, the authors that have high true-positive rates normally repeated at least 10% of their 4-gram characters (corresponding to 8-12 features) per tweet. The recognized tweets do not usually have similar contents. This means that writing styles can be captured by our selected character 4-gram features.

Twitter user	Re-used char 4-grams	Detected percentage
Y	10%	90%
R D	5%	80%
N	6%	100%
A	2%	0%

Table 1: Detailed results of detecting tweet feeds of specific Twitter users (in a pool of 2 candidates).

6.4 Mutual information of features

Feature	Mutual information value
to	0.2981
inde	0.2981
or p	0.2981
*rea	0.2770
yzcq	0.2770
de l	0.2770
//t.	0.2770
q5cb	0.2762
to w	0.2762
! 1	0.2762
was	0.2762
terr	0.2762

Table 2: Mutual information values of features.

For each single tweet in the test phase and a tweet set in the training phase, we extracted features which are then transformed in to a high dimensional vector, one for each phase. These vectors are the inputs for our classifier. The features in these vectors can be considered equally important. Figure 2 demonstrates our features in these vectors with their mutual information values.

6.5 Trump's tweets and his fraudulent accounts

We considered a real work application of our algorithm to determine how effectively our algorithm can attribute fake tweets of a celebrity account. We looked for parody accounts of US's president Donald Trump and his very own account (@realDonaldTrump). We assumed that he writes most of his tweets. There is chance that his tweets are written by his assistants, however, he is known for writing his own tweets, thus we assumed this number is relatively small in the total number of tweets and affects negligibly to our extracted stylometric feature sets.

He has a number of parody accounts and some of them state that they are parody account themselves (@realDonaldTrump), the others are not (@ArtHouseTrump, RealTrump2016). However for the research purpose we tested all of these accounts. Tweets in these accounts are separated to simulate the case when only single tweets are posted and if our algorithm can detect them. We removed all retweets and tweets which are replicated from the @realDonaldTrump account.

Twitter account	Number of tweets	Detection accuracy
@realDonaldTrump	21000	N/A
@ArtHouseTrump	316	99%
@realDonaldTrump	1131	100%
@RealTrump2016	771	100%

Table 3: A description of different Donald Trump's accounts and our fraudulent tweet detection results.

The results are shown in Table 3, which is not surprising as we have a large set of training tweets from the real Donald Trump account and the algorithm only answers yes or no to a tweet.

7 SUMMARY AND CONCLUSIONS

We have shown that authors of very short texts can be identified in flexible settings where we have to rely on only a single tweet to construct the user model. We presented the authorship attribution algorithm on micro-messages that uses flexible features and effective feature extraction method - and demonstrated that combining these factors significantly improves the accuracy.

In these scenarios, fake tweets can be detected and authors of anonymous tweets or of a shared Twitter account can be shown. This also has other uses, such as verifying the source of illegitimate tweets. Collecting scam and abusive tweet messages and a set of candidate authors to detect the ownership of these tweets is an interesting extension of this work. In contrast to scam and abusive messages which are available, the set of candidates such as the accounts with pre-history of sending these tweets, is not published or still difficult to collect.

APPENDIX

Below is a list of the function words and other words that we consider to be reused repetitively by the authors and are somehow independent of the contexts. Different lists of function words available online are merged to obtain our list (1). The second list (2) was created by selecting the words used multiple times by the authors in our corpus regarding their writing topics.

(1) List of function words: *a, able, aboard, about, above, absent, according, accordingly, across, after, against, ahead, albeit, all, along, alongside, although, am, amid, amidst, among, amongst, amount, an, and, another, anti, any, anybody, anyone, anything, are, around, as, aside, astraddle, astride, at, away, bar, barring, be, because, been, before, behind, being, below, beneath, beside, besides, better, between, beyond, bit, both, but, by, can, certain, circa, close, concerning, consequently, considering, could, couple, dare, deal, despite, down, due, during, each, eight, eighth, either, enough, every, everybody, everyone, everything, except, excepting, excluding, failing, few, fewer, fifth, first, five, following, for, four, fourth, from, front, given, good, great, had, half, have, he, heaps, hence, her, hers, herself, him, himself, his, however, i, if, in, including, inside, instead, into, is, it, its, itself, keeping, lack, less, like, little, loads, lots, majority, many, masses, may, me, might, mine, minority, minus, more, most, much, must, my, myself, near, need, neither, nevertheless, next, nine, ninth, no, nobody, none, nor, nothing, notwithstanding, number, numbers, of, off, on, once, one, onto, opposite, or, other, ought, our, ours, ourselves, out, outside, over, part, past, pending, per, pertaining, place, plenty, plethora, plus, quantities, quantity, quarter, regarding, remainder, respecting, rest, round, save, saving, second, seven, seventh, several, shall, she, should, similar, since, six, sixth, so, some, somebody, someone, something, spite, such, ten, tenth, than, thanks, that, the, their, theirs, them, themselves, then, thence, therefore, these, they, third, this, those, though, three, through, throughout, thru, thus, till, time, to, tons, top, toward, towards, two, under, underneath, unless, unlike, until, unto, up, upon, us, used, various, versus, via, view, wanting, was, we, were, what, whatever, when, whenever, where, whereas, wherever, whether, which, whichever, while, whilst, who, whoever, whole, whom, whomever, whose, will, with, within, without, would, yet, you, your, yours, yourself, yourselves*

(2) Example of other words: *buy, make, option, catch, gather, concern, look, see, right, consider, strategy, company, rush, stop, circumvent, try, american, phone, adult, embark, find, associate ...*

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